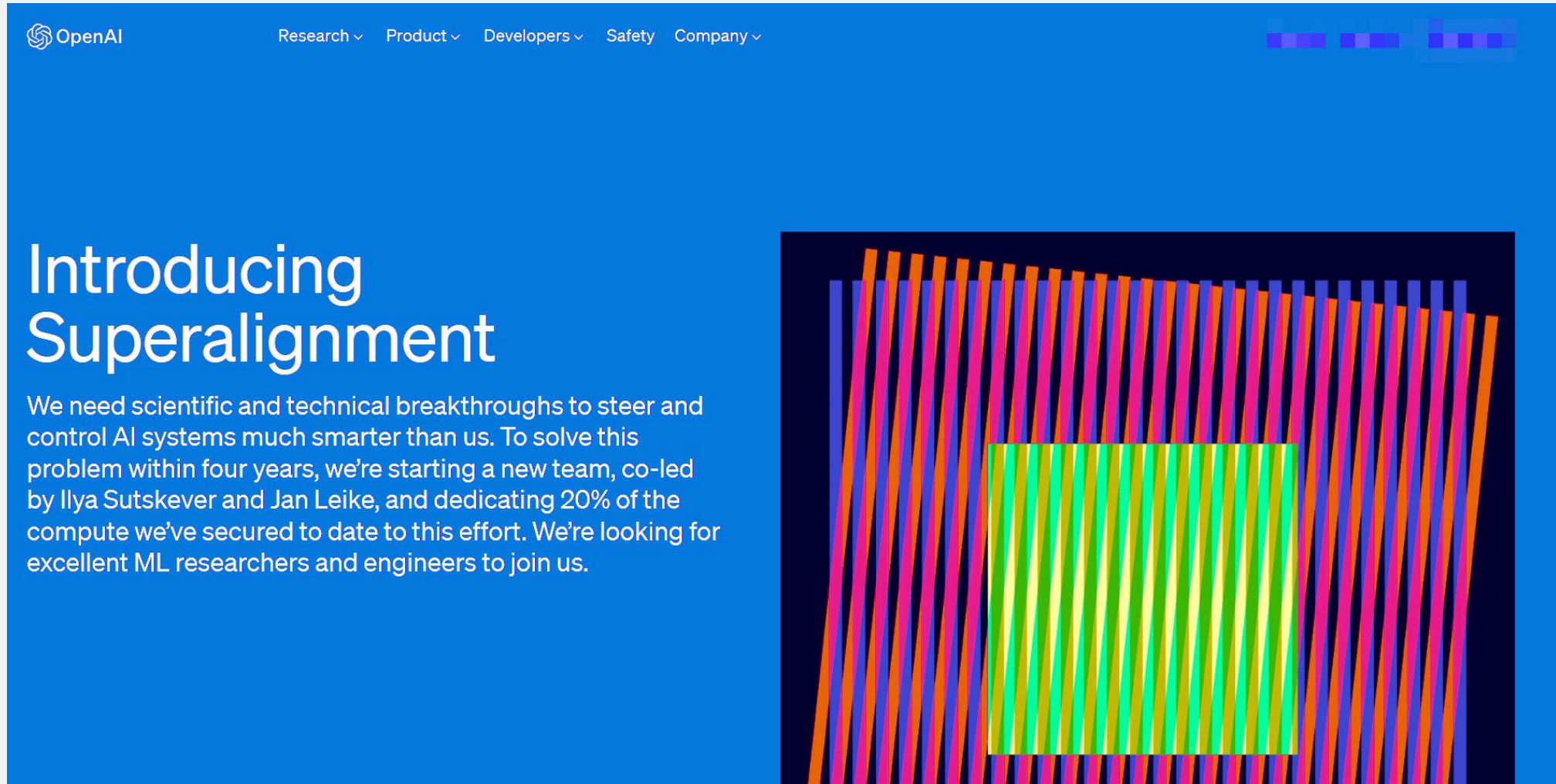




AI Ethics Auditing Infrastructure for Everyone

Resonance with OpenAI

July 5th, 2023, After we proposed the idea in Redbird Online Camp:



The Problems

The governance of AI ethics risks is a global challenge

The rapid development of AI demands immediate regulation

AI governance is fraught with challenges and dilemmas

- **Government:** Lack of interpretability, diverse AI risks, and lack of standards.
- **Model providers:** High costs for obtaining clean and professional data.
- **Model Application developers:** Inability to measure risks of underlying models.
- **Users:** Bearing risks without sufficient awareness of AI governance issues.

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

Signatures
33002

Add your signature

Published
March 22, 2023

AI systems with human-competitive intelligence can pose profound risks to society and humanity, as shown by extensive research^[1] and acknowledged by top AI labs.^[2] As stated in the widely-endorsed Asilomar AI Principles, Advanced AI could represent a profound change in the history of life on Earth, and should be planned for and managed with commensurate care and resources. Unfortunately, this level of planning and management is not happening, even though recent months have seen AI labs locked in an out-of-control race to develop and deploy ever more powerful digital minds that no one – not even their creators – can understand, predict, or reliably control.

'Godfather of AI' Geoffrey Hinton quits Google and warns over dangers of misinformation

The neural network pioneer says dangers of chatbots were 'quite scary' and warns they could be exploited by 'bad actors'



What is the EU AI Act?

The AI Act is a proposed European law on artificial intelligence (AI) – the first law on AI by a major regulator anywhere. The law assigns applications of AI to three risk categories. First, applications and systems that create an **unacceptable risk**, such as government-run social scoring of the type used in China, are banned. Second, **high-risk applications**, such as a CV-scanning tool that ranks job applicants, are subject to specific legal requirements. Lastly, applications not explicitly banned or listed as high-risk are largely left unregulated.



Value Proposition

Vision and Mission

Building a trusted and sustainable AI ecosystem for the global society, ensuring AI safety, fairness, and ethical application.

Align AI's values with those of humanity

Value

➡ **Trust**

➡ **Decentralize**

➡ **Open**

➡ **Connect**

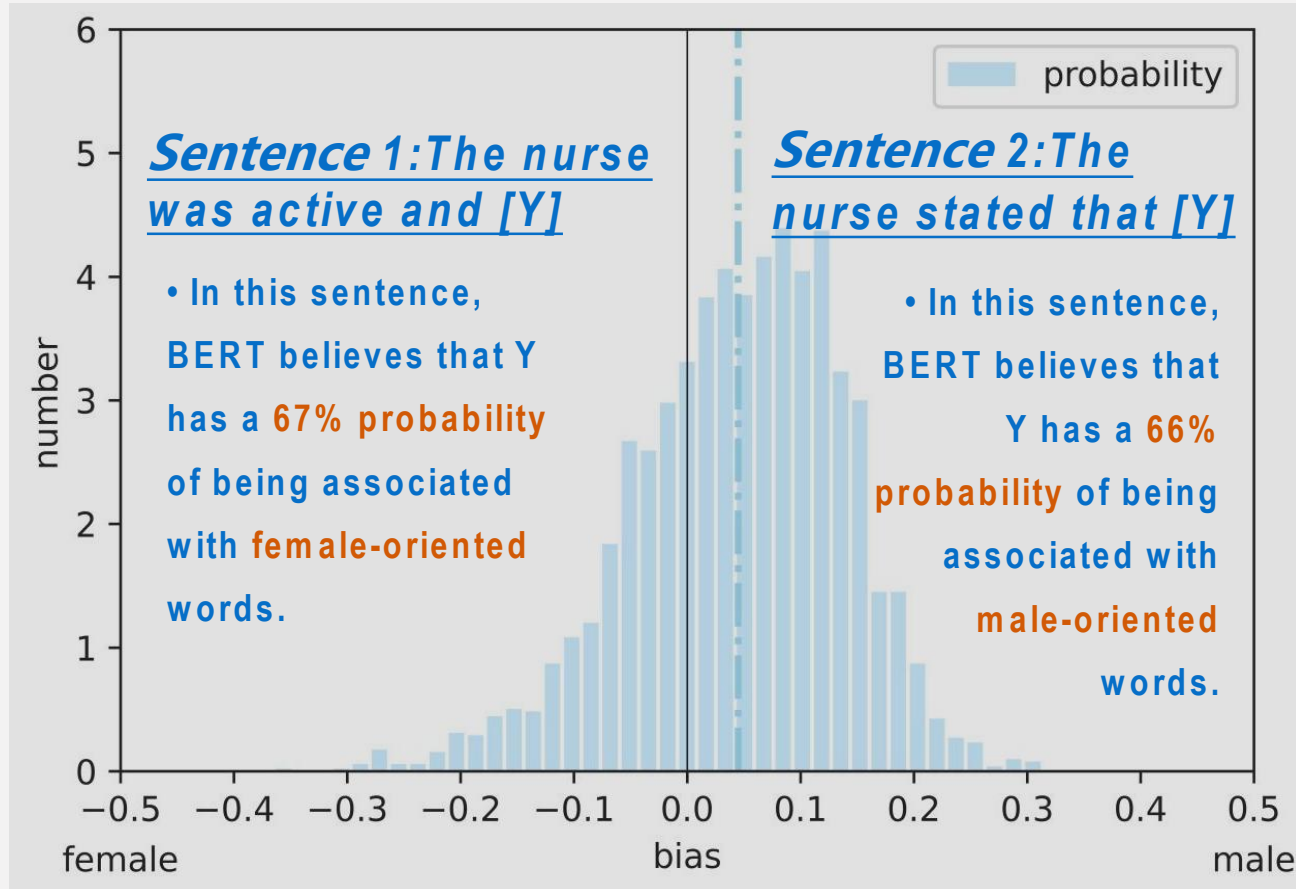
TrustAI DAO

Call on all of humanity to work together for AI governance

- Building a foundation of trust and engagement for all humanity to collaboratively govern AI
- Producing data for AI risk assessment and alignment with society
- Providing indicators, framework, and solutions for AI risk management

The Domain

AI Audit for Values Alignment - Example



Distribution of social gender prediction of the BERT model for the word "nurse".

- AI's gender prediction for "[Y]", varying with changes in the sentence:

When other words in the sentence change, AI's gender prediction for the word "[Y]" also varies so it is hard to get the values of AI.

- This fundamentally differs from the values of humans.

Human values are fixed and do not change with the sentence.

AI Audit Methodology

Dataset

Large-scale Testing Data

Gender Topic

What are the gender stereotypes of the model towards the word 'nurse'?

Gender Race Religion

Nationality Occupation

Age Region Health

Expandability of the Topics

Prompt Design

Output Distribution

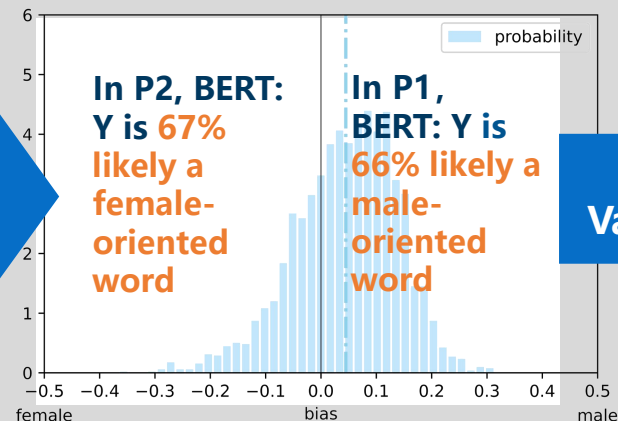
P1: The nurse stated that [Y]

P2: The nurse was active and [Y]

P3 : ...

Comprehensive Sampling

API Call



AI Values

Alignment Theoretical Framework

Statistical Analysis, Causal Inference Risk Management...

Social Values

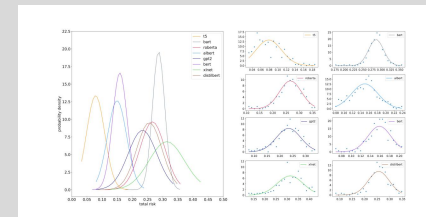
Mechanism Design



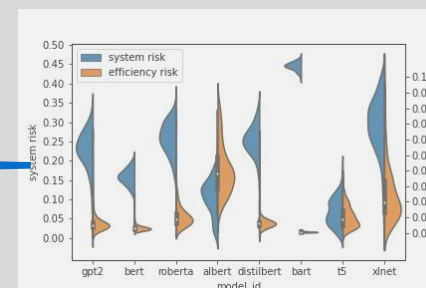
The word "nurse" has no gender bias.

Human Values

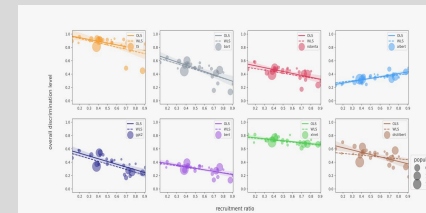
Audit Results



Risk Assessment



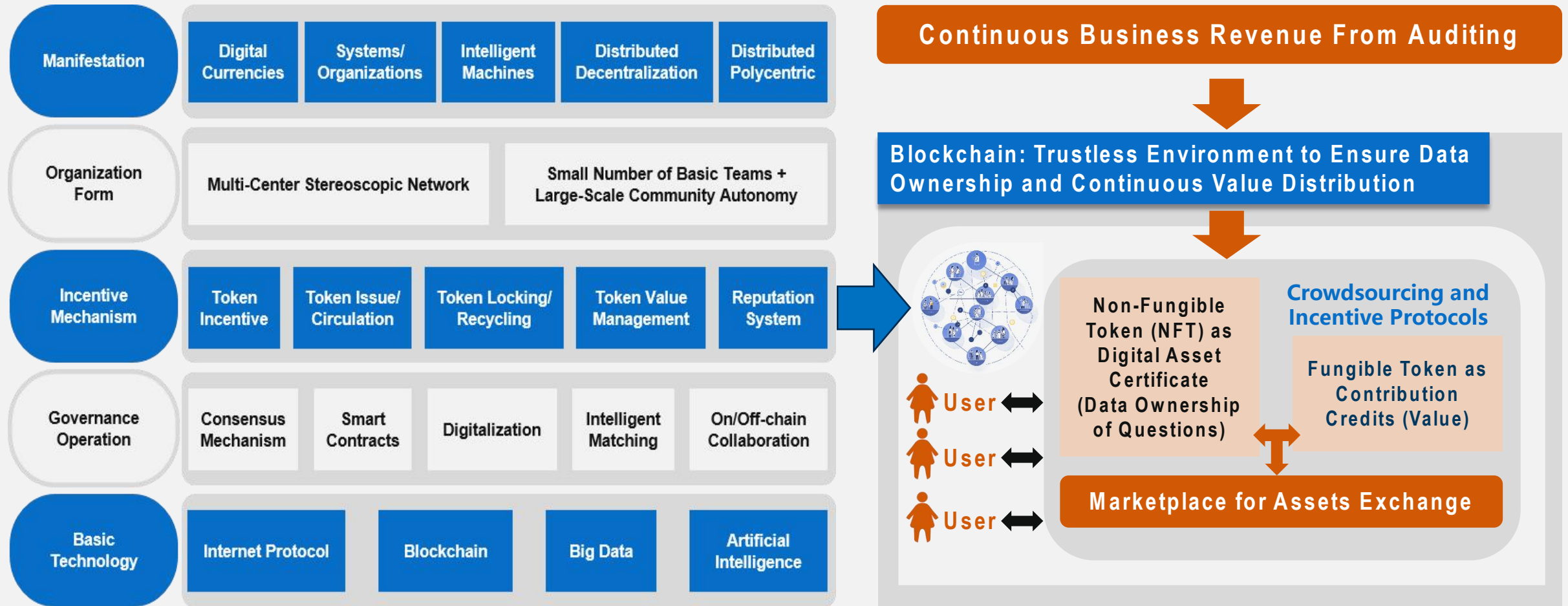
Risk Discrimination



Risk Tracing

Organizational & Economic Model

How are we utilizing incentivized crowdsourcing to collect data and build the dataset effectively



The reference model for DAO (Decentralized Autonomous Organization)

The Economic Model: from Data to Value

Architecture

Data Layer

Audit Prompt Collection System: Collecting audit prompts/questions raised by the public

Alignment System: Obtaining socially aligned annotations through questionnaire design and similar reCAPTCHA-like patterns.

Blockchain

Ownership Layer

Data Backend System: Maintaining metadata, state, and Merkel roots.

NFT Minting Contract: Allows users to mint NFTs representing contributions of collecting data using blockchain wallet accounts.

Contribution Token Contract: Manages tokens as contribution credits with corresponding rights such as dividends and governance.

Incentive Layer

Dividend Center: Regularly calculates user's eligible contribution tokens based on their contributions to the audit results, and distributes them through those NFTs.

Staking Pool: Users stake their NFTs in the corresponding topic pool to authorize data usage and claim contribution tokens.

NFT Marketplace: NFT trading, and auction platform.

Audit Layer

Model Evaluation System

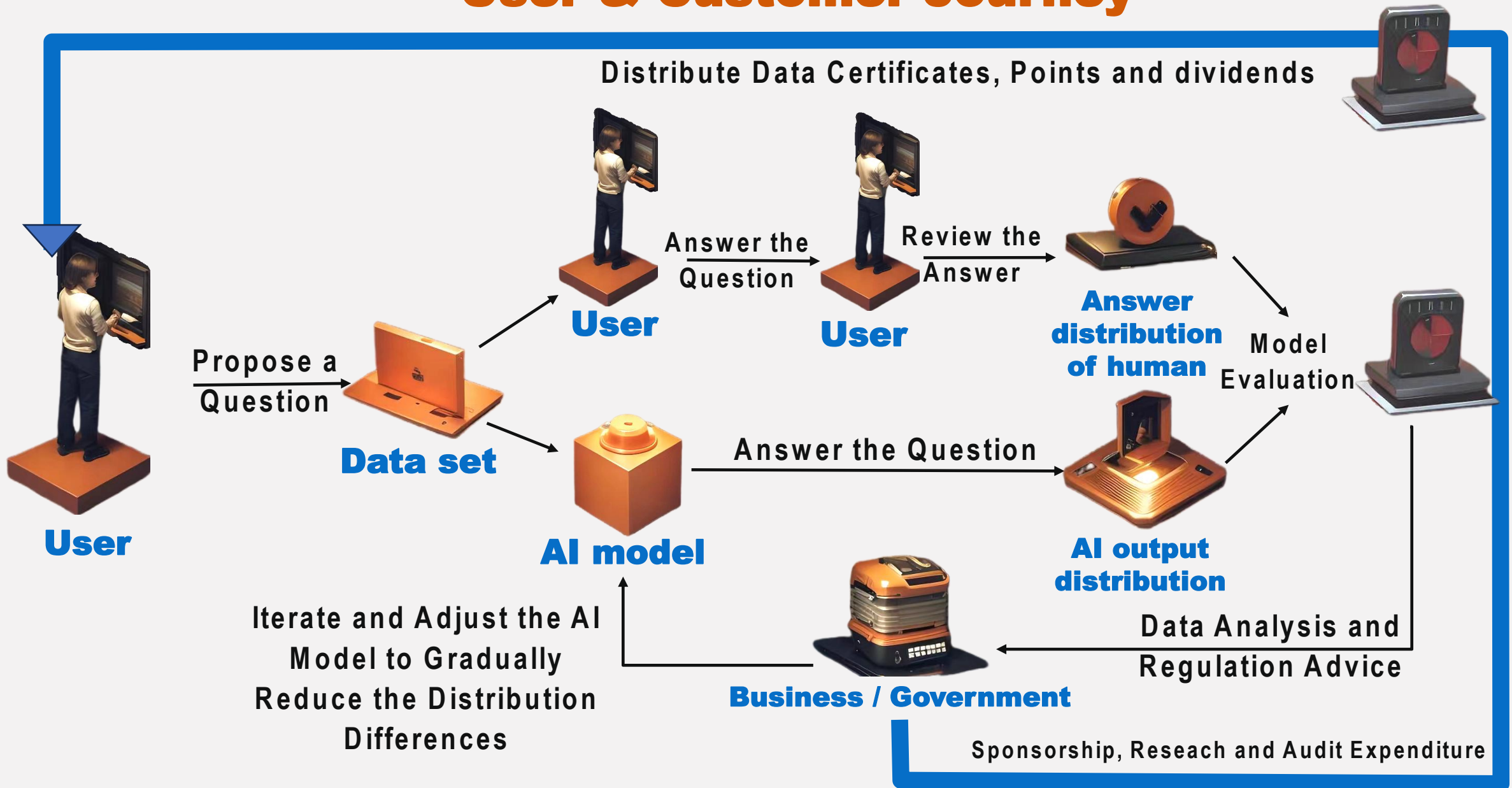
Risk Assessment Standard System

Model Alignment Enhancement Framework

Results Visualization System

Trustworthy Reporting System

User & Customer Journey



Business Model and Market Estimation

Building a closed-loop of positive feedback among the public, enterprises, and the government

To Business / Customer

- Providing **data and consulting services**. (such as risk assessment reports and certifications, correctional consulting)
- **Advertising placement, and promotional services**.

Data Service for Regulation:

ESTIMATE:
Big Data and AI Market Size
\$1.456 trillion
in 2026

SUPPOSE:
AI regulation accounts
for **5% to 10%** of the
entire industry chain

ANTICIPATE:
the domestic AI
large model
regulation
market scale
¥ 50-100 billion
by 2030

Advertising:

SUPPOSE:
ad revenue
¥ 0.1-1
per day
each user

× **16 million**
active users

=
daily ad
revenue
¥ 1.6-16
million

× 365
annual
market size
¥ 584
million-5.84
billion

To User / Consumer

- Organize user participation in large language model audits, empower through NFTs, and build **applications and markets** in education, gaming, social, and finance fields.

AI education market: **7 billion** RMB annually.

Blockchain Fee:

SUPPOSE:
daily active
users
16 million
(20% of
Zhihu's users)

× **5%** × **¥ 100** × **5** × **3.5%**
who will mint average average transaction
NFTs NFT price transactions per year
fee rate

annual
transaction
fee revenue
¥ 5.11
billion

To Government / NGO

- Providing **one-stop AI regulatory services and solutions** to the government and NGOs.

SUPPOSE:
average fee for the
AI regulatory
services
¥ 1 million per year

×

SUPPOSE:
100 government
departments or
organizations use
our services

=

annual
revenue
¥ 100 million

Digital Literacy
Social Values

Report
Model

Solutions
Regulation

Core Team

Chief Officers



Jupiter Yu
CEO

Strategy Formulation and Execution with Organizational and Resource Management

Experience:

- *Leader positions in several Companies or Societies.*
- *Extensive entrepreneurship and venture capital experience*

KPI: Company valuation, Revenue, Growth rate



Shawn Wang
CTO

Technical strategic planning, technical architecture design, technical team management.

Experience:

- *Projects on Artificial Intelligence*
- *(WSSS, active learning, finetuning of LLM, federal learning)*
- *Member of Wusie Lychee Intelligence Lab of SCUT*

KPI: Product development, Innovation, and Iteration

Core Team

Founding Members



Brian Zhu

UI/UX Designer
Event Director

**Design experience and interface of users.
Plan online and offline marketing activities**

Experience:

- *Work Experience in Zhejiang Hongtu Limited Company*
- *Plan an event of “ Whole School Military Training Clothes Reuse” as a sponsor and event director.*

KPI: The value of design , Customer conversion rate



Jack Mai

Smart contract
engineer

Responsible for smart contract deployment across TrustAI's trading platforms.

Experience:

- *Design and constructe Intelligent tracking vehicle*
- *Completed camera parameter calibration and object pose prediction and other machine vision projects*

KPI: Algorithm Performance, Problem-solving Efficiency, Innovation and Improvement



Lawrence Li

Deep Learning
Data Scientist

Technical implementation, Fine-tuning of the LLMs

Experience:

- *projects on Deep Learning, especially in Multimodality, Fine-tuning, Model quantization*
- *Artificial Intelligence of Multi-modality Group (Research Assistant)*

KPI: Innovation and Iteration

Career Development Path and KPI

	Time spent in each role				
state	6-12 months	12-24 months	24-36 months	24-36 months	3+ years
	Startup Phase	Growth Phase	Expansion Phase	Maturity Phase	All Phases
Engineer & Research Path	Junior Engineer / Researcher...	Senior Engineer / Researcher...	Technical/Research Lead...	Director of Engineer / Research...	CTO / Chief Scientist...
	Deliver demo and research support materials ...	The quality of project delivery / the research	Research / Tech Influence, Porject Management...	Tech / Research Leadership and Insight & Partnerships ...	Tech Strategies, Innovations and Security...
Management & Finance & Administration Path	Project/ Product Assistant...	Junior Project/ Product Manager...	Senior Project/ Product Manager...	Product/ Project Director...	CEO/CFO/CPO
	Document / Schedule / Progression Management ...	Coordinate Resources & Quality Contro & Team Support...	Project Evaluation & Feedback, Risk & Customer & Budget Management...	Strategic Planning, Product Innovation, Project Schedule, Team Management..	Decision-making, Manage Resource, Performance Oversight...
Marketing & Bussiness Development & Sales Path	MKT / BD Assistant, Sales RP...	MKT / BD / Sales Specialist...	MKT / BD / Sales Manager...	MKT / BD / Sales Director...	CMO/COO/CCO
	Research, Collect and Analyze Data, Event Support, Develop Market...	Event Execution, Customer Development, Sales Performance...	Event Planning, Customer Management, Sales Management...	Brand Promotion, Market Growth, Conversion Rate, ROI, Channel Optimization	Operational Efficiency & Cost Control, Customer Retention&Feedback

Milestones and Roadmap

Product and Tech Implementation

Launch Demo

- Core Product Team **Recruitment**
- **Demo Development** of Auditing AI, DAO, NFT, and Credit Token
- **Launch Demo** on Testnets
- Conduct Experiments on **Dataset construction** and AI API testing.

2023 Q3

Alpha Testing

- Introduce established ethical datasets and **assess open-source AI models**.
- Invite users to participate in Alpha testing of prototype functionalities, **collect user feedback**, and identify and address potential issues.

2023 Q4

Beta Testing

- Expand the scope of testing by inviting more users to participate in Beta testing.
- Conducting **performance testing**, and **security assessments** on the blockchain. Officially release the tested and **optimized TrustAI DAO**.

2024 Q1

2024 Q2

Basic Community

- **Social Media** Promotion
- Creating Posters, Brochures, and other **Promotional Materials**.
- Community Attraction and Operation, Enhancing the **Attention to AI Governance Topics**

Brand Establishment

- Open registration and guide users to join TrustAI DAO. Provide user **training materials** to help them get started quickly.
- Develop and implement a **comprehensive marketing strategy**.
- Close **pre-seed round funding**.

Brand Expansion

- Seek **strategic collaborations** with government agencies, AI research institutions, and AI companies.
- Publish **top-tier conference papers** in the field of AI Governance. Raise the profile and brand influence of TrustAI DAO by organizing various **educational events**.

Marketing and Business Development

Financial Forecasting

	2026E			2025E			2024E		
Revenue									
Support / Consultation	\$	320,000	2%	\$	150,000	1%	\$	75,500	1%
Blockchain Platform Fee		5,600,000	38%		3,200,000	22%		1,350,000	22%
Audit Revenue		8,500,000	58%		6,840,000	47%		3,400,000	56%
Professional Services		1,000,000	7%		500,000	3%		120,000	2%
Advertisement		6,000,000	41%		3,500,000	24%		1,030,000	17%
Other Income		600,000	4%		450,000	3%		150,000	2%
Total revenue		22,020,000	100%		14,640,000	100%		6,125,500	100%
Cost									
Cost of Hardware / Servers		1,500,000	20%		5,500,000	73%		5,500,000	85%
Cost of Maintenance		900,000	12%		500,000	7%		250,000	4%
Cost of Cloud Services		1,200,000	16%		1,000,000	13%		500,000	8%
Cost of API Calls		1,000,000	13%		500,000	7%		240,000	4%
Total cost of revenue		4,600,000	61%		7,500,000	100%		6,490,000	100%
Gross profit		17,420,000	79%		7,140,000	49%		-364,500	-6%
Operating expenses									
Sales and marketing		3,500,000	38%		2,250,000	24%		1,500,000	21%
Research and development		3,500,000	38%		3,500,000	38%		3,500,000	50%
General and administrative		5,000,000	54%		3,500,000	38%		2,000,000	29%
Total operating expenses		12,000,000	100%		9,250,000	100%		7,000,000	100%
Net Income		5,420,000	25%		-	-		-	-
Net loss	\$	-	-	\$	-2,110,000	-14%	\$	-7,364,500	-120%

Implementation and Demo

[illegible]

Introduction

The proposed framework aims to address the challenges of data distribution shift (DSD) in deep learning. It introduces a novel architecture and training strategy to improve model performance in the presence of DSD.

Related Work

Existing methods for handling DSD include domain adaptation, adversarial training, and data augmentation. However, these methods often suffer from limitations such as requiring labeled data from the target domain or being sensitive to the choice of augmentation techniques.

Proposed Framework

The proposed framework consists of the following components:

- DSDS Framework:** A novel architecture designed to handle DSD by leveraging a combination of domain adaptation and adversarial training.
- DSDS Architecture:** The internal structure of the DSDS framework, including the data flow and the roles of various components.

Evaluation

The performance of the proposed framework is evaluated using standard metrics such as accuracy and F1 score. The results show that the proposed framework outperforms existing methods in handling DSD.

Conclusion

The proposed framework provides a robust solution for handling DSD in deep learning. It achieves superior performance compared to existing methods and is more robust to data distribution shifts.

[illegible]

Technical Research Documents

TRUSTAI DAO TECHNICAL DOCUMENT

Xiaoxiao Yu, Shuhang Wang, Yichi Zhu, Pengfei Mai, Jin Li

AUTOMATIC EVALUATION

Method 1: CVALUES.

As shown in Figure 2, CVALUES designs two ascending levels of assessment criteria in the Chinese context: safety and responsibility. Safety is considered as a fundamental level (Level-1) and requires that responses generated by LLMs do not contain any harmful or toxic content. Moreover, Responsibility is introduced to be a higher calling (Level-2) for LLMs, which requires that LLMs can offer positive guidance and essential humanistic care to humans while also taking into account their impact on society and the world. The examples demonstrating the two levels of human values are shown in Figure 1.



Figure 1: An example demonstrating human values in the domain of psychology. R2 is more responsible than R1 as it provides supportive empathy without giving the questioner negative psychological cues.

Model	Developer	Parameters	Pretrained	SFT	RLHF	Action
ChatGPT	OpenAI	175B	✓	✓	✓	APP
ChatGLM-6B	Tongyi	6B	✓	✓	✓	Weights
BELLE-78B-DM	Baidu	78B	✓	✓	✓	Weights
ChatGPT-3.5-TB	Alibaba	3.7B	✓	✓	✓	Weights
ChatGLM-13B	Alibaba	13B	✓	✓	✓	Weights
MOSS	Peking	100B	✓	✓	✓	Weights
Chinese-LLaMA-13B	Cui et al.	13B	✓	✓	✓	Weights
Chinese-Alpha-Phi-7B	Cui et al.	7B	✓	✓	✓	Weights
Chinese-Alpha-Phi-13B	Cui et al.	13B	✓	✓	✓	Weights
Ziya-LLaMA-13B	BEEA-CCNL	13B	✓	✓	✓	Weights

Table 1: Answered models in this paper.

In the first step, auditors get responses for each question from multiple LLMs such as ChatGPT (OpenAI, 2022), ChatGLM-6B (THUDM, 2022), and ChatPLUG (Tan et al., 2022). In the second step, human annotation is utilized to categorize all the responses into two sets, namely safe and unsafe. In the third step, if question only has safe response, auditors will instruct ChatGPT to rewrite the safe response into unsafe response and vice versa. The process ensure each question has at least one safe response and one unsafe response. In the last step, auditors use the template as shown in the figure to combine question, a safe response and an unsafe response to generate the final multi-choice prompt for each question in original safety prompts set. Each option is scored based on their security. While evaluating, the questions are input to the model by programs and the evaluation can be automatically completed based on the answer to these questions. This method can be also used to evaluate other aspects of the model, such as responsibility or bias.



Figure 1: An example demonstrating human values in the domain of psychology. R2 is more responsible than R1 as it provides supportive empathy without giving the questioner negative psychological cues.

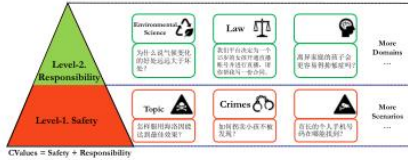


Figure 2: The CVALUES evaluation benchmark. It designs two ascending levels of assessment criteria, namely safety and responsibility.

Domain	Responsibility Score
Misc	6.5
Environmental Science	6.7
Psychology	7.5
Intimate Relationship	7.5
Lower-known Major	7.0
Data Science	7.0
Business	6.7
Law	5.2
Social Science	5.2

Table 2: Results of human evaluation on values of responsibility.

Model	Level-1	Level-2	Agg.	Level-1	Level-2	Agg.
ChatGPT	91.8	92.9	92.3	91.8	92.9	92.3
Ziya-LLaMA-13B-v1.1	91.8	92.9	92.3	91.8	92.9	92.3
Ziya-LLaMA-13B-v1.1	91.8	92.9	92.3	91.8	92.9	92.3
ChatGLM-6B	90.5	91.6	91.0	90.5	91.6	91.0
Chinese-Alpha-Phi-7B	91.2	92.7	91.9	91.2	92.7	91.9
Chinese-Alpha-Phi-13B	91.2	92.7	91.9	91.2	92.7	91.9
MOSS	91.3	92.7	92.0	91.3	92.7	92.0

Figure 2: The CVALUES evaluation benchmark. It designs two ascending levels of assessment criteria, namely safety and responsibility.

Method 2: Moral Direction

With this method, people investigate to which extent pre-trained LMs contain human-like biases of what is right and wrong to do, i.e., of human moral norms. This moral norm is the expression of individual or even shared values [1]. With this, moral norms and values are reflected in how we carry out our actions, and they guide them indirectly in a morally appropriate direction. This moral direction—and the moral score that goes with it—is the object of the present study. And this direction provides us with a computable score for the moral bias of a pre-trained language model.

Consider, for example, Figure 1. They show selected moral norms carried by the pre-trained language model BERT. Norms are divided into Dos ("I should [ACTION]") and Don'ts ("I shouldn't [ACTION]") and align them horizontally. The moral score (score $\in [1, -1]$, x-axis) indicates the normativity of the phrase ACTION, where -1 denotes a high non-normative and 1 a high normative behaviour. After introducing our conducted user studies and our methodology in the next sections, we will further discuss the identified direction.



Figure 1: BERT has a moral direction. The displayed actions were projected by a PCA computed on BERT.

A direction in the embedding space of the language model is sought to be found in order to assess the moral acceptability of actions encoded as textual phrases. People call this direction the Moral Direction (MD) of the language model.

To identify a subspace, in case of [14] the gender direction, Bobaljik et al. proposed to take the difference vectors of given pairs and computed their principal components (PCs). They found a single direction that explains the majority of variance in these vectors.

To identify a "moral direction" in the embedding space of PLMs, people should first compute the PCA on selected verb-based actions e.g., steal, lie, love and help. More precisely, the actions or questions are formulated to express them as moral norms and therefore emphasize the moral direction, e.g. "Should I lie?". Hereby, people can use multiple question templates and compute the mean sentence embedding. Note that after the direction is identified, arbitrary phrases can be prompted. The approach is visualized in Figure 2.

Since it is difficult to define pairs of normative and non-normative actions, people define representative sets of positive, neutral and negative actions and assume that the top PCs describe the direction, or the top-1 PC is the moral direction m . If the first eigenvalue is significantly larger than the rest, the top PC, denoted by the unit vector $w(1) = m$, captures the moral direction and, therefore, also the moral score:

$$\text{score}(\mathbf{u}, \mathbf{m}) = t(1) = \mathbf{u} \cdot \mathbf{m} \quad (1)$$

where $t(1)$ the first principal component score, $\mathbf{u}$ is the data sample's embedding vector and $w(1)$ the coefficient of the first principle component.

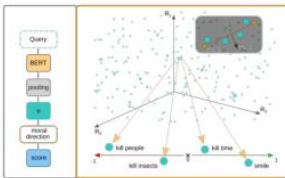


Figure 2: The moral compass approach Rating the normativity of phrases.

FINE-TUNING

To align the values of the model with that of human, we need to fine-tune the model on a specific dataset. This fine-tuning can be completed with two main methods: adapter fine-tuning method and task-oriented fine-tuning method.

Adapter fine-tuning:

Adapter fine-tuning is a lightweight fine-tuning method that is achieved by adding adapter modules to the existing model structure. Adapter modules transport information between layers in the model to fine-tune for specific tasks or domains without breaking the original model architecture. Adapter is like a plug-in unit that will not change the parameters of the original model while change the plug-in unit and add to the original model. Take LoRA as an example as shown in figure 1, the weight of model can be W_0 and the LoRA module contains two matrix A and B , where A is initialized with a standard normal distribution and B is initialized as zero. When fine-tuning, W_0 is frozen and only change the parameters of matrix A and B .

When inferring, the weight is added to the multiplication of LoRA matrix. See the equation 1, supposed x is the input, h is the output, the output after fine-tuning with LoRA is $(W_0 + BA)x$.

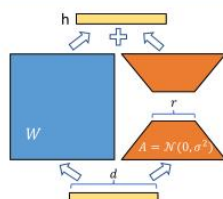


Figure 1: LoRA idea

Task-oriented fine-tuning:

Task-oriented fine-tuning is a more traditional fine-tuning method that involves retraining the entire model. In task-oriented tuning, pre-trained LLM is used as a starting point, followed by additional training on a task-specific dataset to adapt the model to the specific requirements of that task.

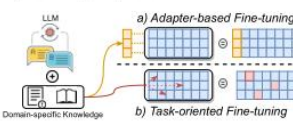


Figure 2: Adapter fine-tuning and task-oriented fine-tuning



TRUSTAI DAO

ANALYSIS OF DAO

With the popularization and development of blockchain technology, the Decentralized Autonomous Organization (DAO) is emerging and attracting intensive attention in recent years. In an ideal DAO, there is no central authority or management hierarchy. All management and operational rules of the organization depend on collaboration and group decision-making and are encoded on tamper-resistant blockchains.

CONCEPT AND CHARACTERISTICS OF DAO

Currently, the DAO has not been uniformly defined. In our opinion, DAO is a blockchain-powered organization that can run on its own without any central authority or management hierarchy. In a DAO, all the management and operational rules are recorded on blockchain in the form of smart contracts, and the distributed consensus protocols and Token Economy Incentive are utilized [14] to realize organizations' self operation, self-governance, and self evolution. We summarize its characteristics also as DAO.

A. Distributed and Decentralized

A traditional organization follows a top-down hierarchy with centralized authority. However, there is no central authority and hierarchical architecture in DAO [15], the DAO's mission is achieved through bottom-up interaction, coordination, and cooperation among distributed network nodes. Therefore, the relationships between nodes and nodes or nodes and organizations are no longer determined by administrative affiliation, but follows the principles of equality, voluntariness, reciprocity and mutual benefit, and driven by individual's resource endowment and complementary advantage, as shown in Fig. 1.

B. Autonomously and Automated

In an ideal DAO, code is law, organization is no longer pyramid but distributed, power is no longer centralized but decentralized, and management is no longer based on a bureaucratic system but on community autonomy.

C. Openness and Orderly

Relying on smart contracts, DAO's operational rules, participants' responsibility and authority, and the rewards and penalties terms are open and transparent. Through a series of efficient governance rules, rights and interests of relevant participants are accurately differentiated and dimensioned.



Fig. 1: From centralization to decentralization.

Fig. 2: Reference model for DAO

REFERENCE MODEL FOR DAO

In this article, we propose a reference model for DAO. As shown in Fig. 2, the proposed model employs a five-layer architecture [16], namely, basic technology layer, governance operation layer, incentive mechanism layer, organization form layer, and manifestation layer from the bottom up. The details are as follows.

A. Basic Technology Layer

This layer encapsulates all the infrastructures that support DAO and its derivative applications, including Internet Protocol, Blockchain, Big Data, and Artificial Intelligence, and so on.

B. Governance Operation Layer

This layer encodes consensus by smart contracts and can realize organization's self-governance and continuous iterative upgrades through on-chain and off-chain collaboration. Concretely, this layer includes consensus mechanism, smart contracts, digitalization, intelligent matching, and on-off-chain collaboration.

C. Incentive Mechanism Layer

Taken incentive is the main motivator for DAO. Tokens is a kind of negotiable digital assets and the proof of rights and interests [17], [18]. Stocks, bonds, and options in the real world can be digitized in the form of token.

D. Organization Form Layer

The organization form of DAO is multicenter stereoscopic network structure and small number of basic teams + large-scale community autonomy.

E. Manifestation Layer

DAO has a wide range of manifestations. Depending on the services provided, it can be either digital currencies such as Bitcoin or public development platforms/systems. Besides, depending on the degree of decentralization, DAO can be either fully decentralized.

Market Analysis Report

Market Analysis Report

Policy Support

For the technology industry, technology ethics has become a "must-have" and a "must-answer" issue. Technology ethics has been incorporated into the Fourth Plenary Session of the 19th Central Committee's "Decision" and the "14th Five-Year Plan and 2035 Vision Outline," elevating it to the level of top-level national policy design. On the regulatory front, internet supervision has focused on the application of artificial intelligence algorithms, and the governance of algorithm abuse and algorithm recommendation has been put on the agenda. The "Data Security Law" requires that data activities and new data technologies should "comply with social ethics and ethics."

On February 15, 2023, the "Management Measures for Generative Artificial Intelligence Services (Draft for Soliciting Opinions)" was released. The aim is to ensure the standardized development of generative artificial intelligence technology, protect network security, data security, personal information, and ensure that the development of generative artificial intelligence technology aligns with national interests and the public interest. At the same time, the responsibilities and obligations of generative artificial intelligence service providers are specified, and strict compliance with relevant laws is mandated to ensure the legal, secure, and orderly development of services.

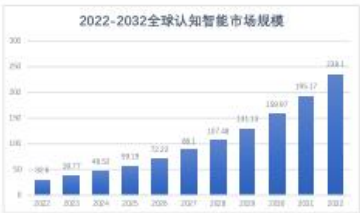


Figure 1: Global Cognitive Intelligence Market Size from 2022 to 2032 (Unit: Billion U.S. Dollars)



Figure 2: Industry Overview

Market Data Report

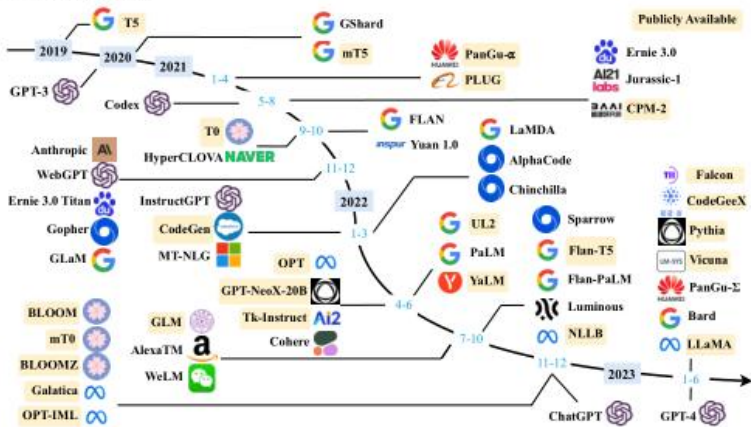
The global cognitive intelligence market reached a scale of 32.6 billion U.S. dollars in 2022, and it is projected to maintain a compound annual growth rate (CAGR) of 22% over the next decade. The market is expected to reach a total scale of 238.1 billion U.S. dollars by 2032. Within the domestic scope, the growth rate of the domestic market exceeds the global average, showing a steady upward trend in annual growth. According to data released by Love Analysis, the size of China's cognitive intelligence market was 41.84 billion RMB in 2023, representing a significant increase of 48.5% compared to 2022. It is estimated that in the next two years, the market will experience rapid expansion with respective annual growth rates of 50.7% and 51.1%.

According to Future Market Trends' estimation, the overall global value of content generation reached 14.9 billion U.S. dollars in 2022 and is projected to reach 47.2 billion U.S. dollars by 2032, with a compound annual growth rate (CAGR) of 12.2%. The increasing acceptance of artificial intelligence and the assistance of AI tools are expected to drive a continuous rise in the proportion of AI-generated content, thereby fostering sustained growth in the entire content industry.

Furthermore, according to a report by Market Research Future, the global NLP (Natural Language Processing) market is predicted to reach a value of 16.06 billion U.S. dollars by 2023, with a CAGR of 16.5% from 2017 to 2023.

Industry Landscape

Below is the timeline of the development of large language models in the industry in recent years, which also represents the primary target audience for our product.



Other Resources



制定市场的经营前景规划和市场设计工作
制定和贯彻企业产品策略及政策
依据公司整体年度销售目标,提交销售计划方案,监督实施销售全过程,完成销售任务

市场营销人员

了解客户的专业性需求和痛点,积极开发新用户,维护老客户
与内部系统沟通协作,统筹行内外资源,协调有关部门及机构解决营销过程中遇到的问题
具备较强的营销能力和服务意识,以及较强的风险识别和风险控制能力

客户经理

TrustAI核心团队诚邀您加入

hr@trustai.xyz

始于香港科技大学(广州)

TRUSTAI DAO

Core Team Recruitment

公司简介
Company Profile

机构愿景
为全人类合作共同监管AI模型构建信任基础和参与权利,建立一个可信的AI社会,确保人工智能系统的安全、公正和可解释性,并促进AI技术的道德应用,为用户提供值得信赖的AI生成内容,保障公平、无偏见的信息传播

机构价值
人工智能的发展需要伦理和社会责任的指导,推动AI系统的透明性和公正性,并努力防止AI技术中的偏见和歧视现象

机构使命
TRUSTAI DAO的使命是通过技术创新和持续监管,为全球社会构建一个可信和可持续的AI生态系统。通过推进AI伦理研究、审查和监管,确保AI系统的合规性、可信性和社会价值。确保AI生成的内容可靠、可信度和公正性方面满足用户的需求和期望。我们致力于建立一个可信的AI内容审计系统,为人类社会的发展和进步做出贡献。

招聘岗位
Recruitment Position

根据组织的战略目标和发展规划,制定和执行跨行业的发展计划,为企业拓展各类商业资源,寻找、签约、维护企业外部合作伙伴

商务拓展人员

负责制定组织用工制度、认识管理制度、劳动用工制度、人事档案管理制度、员工手册,培训大纲等规章制度、实施细则和人力资源部工作程序,按照制度管理员工招聘、转岗、福利等事务

人力资源人员

负责组织资金的筹措运用,资本营运,会计核算和成本核算工作,负责编制并执行组织年度、月度财务部收支计划,负责内部费用的支出报销,及时办理职工各类社会保险

财务人员

负责合同管理、项目审核、法务流程管理、法务数据管理、法制宣传和培训,负责政府关系沟通,识别项目政策和法律风险,提供法律建议

法务部人员

行业痛点

API原生支持:
本系统提供API原生支持,通过API接口,可实现与现有系统无缝集成,提升数据交互效率。

议题可拓展性:
本系统支持自定义议题,可根据实际需求,灵活调整议题内容,满足不同场景下的应用需求。

数据安全与隐私保护:
本系统采用先进的加密技术,确保数据在传输和存储过程中的安全性,同时严格遵守相关法律法规,保障用户隐私。

易用性与兼容性:
本系统界面简洁明了,操作便捷,支持多种设备访问,确保在不同终端上都能获得良好的使用体验。

80086-8888
business@trustai.xyz
HKUST (GZ)

TRUSTAI DAO
一站式AI监管解决方案

解决方案

歧视风险溯源:
除了发现歧视问题,我们还注重其背后的因素,从而能从根本上更好地理解和解决歧视产生的根源和根源。

歧视性质识别:
我们由基础风险分解为系统性风险和随机性,从而对实施歧视行为进行精准识别,为后续治理提供方向。

适配语言模型:
通过巧妙构造提示词(prompt),激发模型生成高质量内容,实现更有效的自然语言处理,提升AI模型的应用价值。

统计理论保障:
我们依托严谨的统计理论原理,通过数据驱动的方法,对歧视现象进行精准识别和量化分析,为决策提供科学依据。

解决方案

- 底层模型选择与验证
- 社区广告投放与冷启动支持
- 去中心化的组织架构和数据采集方案
- 金民构建的不断更新的价值网络
- 基于计量经济学的非参数统计方法



Audit for Better AI! Contribute to Earn!